ADDITIONAL MATERIAL

DANIEL W. COLLISON AND DANIEL TUBBENHAUER

Lemma. *(See [CT26a, Lemma 5A.8] for context). The shifted comultiplication $\Delta^S_{n,\xi}$ satisfies the Frobenius relation and is coassociative.*

Proof. Recall that the Frobenius relation is given by

$$(\mu_{n,\xi}\otimes \mathrm{id})\circ(\mathrm{id}\otimes\Delta^S_{n,\xi}) = \Delta^S_{n,\xi}\circ\mu_{n,\xi} = (\mathrm{id}\otimes\mu_{n,\xi})\circ(\Delta^S_{n,\xi}\otimes\mathrm{id}).$$

The first equality is by construction. Now set $v = x^k y^{k'} \in V_{n,\xi}$, $k\in\{0,1,\dots,n-1\}$, $k'\in\{0,1,2\}$. In case $k'=2$,

$$\begin{aligned}
\big((\mathrm{id}\otimes\mu_{n,\xi})\circ(\Delta^S_{n,\xi}\otimes\mathrm{id})\big)(1\otimes v) &= -\frac{\xi}{3n}\sum_{i+j=1} x^i\otimes x^{j+k}y^2 - \frac{1}{3n}\sum_{i+j=n+1} x^i\otimes x^{j+k}y^2\\
&\quad+\frac{1}{3n}\sum_{i+j=n} x^iy^2\otimes x^{j+k}y^2 + x^iy\otimes x^{j+k+1}y + x^i\otimes x^{j+k+1}y^2\\
&\quad+\frac{\xi}{3n}\Big(y^2\otimes x^ky^2 + y\otimes x^{k+1}y + 1\otimes x^{k+1}y^2\Big)\\
&=\frac{\xi}{3n}\Big(y^2\otimes x^ky^2 + y\otimes x^{k+1}y + x^ky^2\otimes y^2 + x^{k+1}y\otimes y\Big)\\
&\quad+\frac{1}{3n}\sum_{\substack{i+j=n\\ i\neq k}} x^iy^2\otimes x^{j+k}y^2 + \frac{1}{3n}\sum_{\substack{i+j=n\\ i\neq k+1}} x^iy\otimes x^{j+k+1}y\\
&=\frac{\xi}{3n}\Big(y^2\otimes x^ky^2 + y\otimes x^{k+1}y + x^ky^2\otimes y^2 + x^{k+1}y\otimes y\Big)\\
&\quad+\frac{1}{3n}\sum_{\substack{i'+j'=n\\ j'\neq k}} x^{i'+k}y^2\otimes x^{j'}y^2 + \frac{1}{3n}\sum_{\substack{i'+j'=n\\ j'\neq k+1}} x^{i'+k+1}y\otimes x^{j'}y\\
&=\Delta^S_{n,\xi}(v)\\
&=(\Delta^S_{n,\xi}\circ\mu_{n,\xi})(1\otimes v),
\end{aligned}$$

and the cases $k'=0,1$ are similar. Note that if the second equality holds on $1\otimes v$, then it holds for all $v'\otimes v\in V_{n,\xi}\otimes V_{n,\xi}$ since $v'\otimes v = (\mu_{n,\xi}\circ\mathrm{id})\big((v'\otimes 1)\otimes v\big)$ and we can use the first equality to pull multiplication by v' from the left through to the front.

Using the Frobenius relation, to prove coassociativity, we can concentrate on evaluation at 1, the unit, only. Now define pairs of integers (p,r), (p',r'), which are either $(1,0)$ or $(n+1,n)$. The four possible groupings each give a collection of terms in $(\Delta^S_{n,\xi}\otimes\mathrm{id})\circ\Delta^S_{n,\xi}(1)$ of the form

$$\begin{aligned}
&\sum_{\substack{i+j=p\\ k+\ell=p'}} x^{i+k}\otimes x^\ell\otimes x^j + \Big(-\sum_{\substack{i+j=r\\ k+\ell=r'}} x^{i+k}y^2\otimes x^\ell\otimes x^j\Big) + \Big(-\sum_{\substack{i+j=p\\ k+\ell=r'}} x^{i+k}\otimes x^\ell y^2\otimes x^j\Big)\\
&+\sum_{\substack{i+j=r\\ k+\ell=r'}} x^{i+k}y^2\otimes x^\ell y^2\otimes x^j + \sum_{\substack{i+j=r\\ k+\ell=r'}} x^{i+k}y^2\otimes x^\ell y\otimes x^jy + \sum_{\substack{i+j=r\\ k+\ell=r'}} x^{i+k}y\otimes x^\ell y^2\otimes x^jy\\
&+\Big(-\sum_{\substack{i+j=r\\ k+\ell=p'}} x^{i+k}\otimes x^\ell\otimes x^jy^2\Big) + \sum_{\substack{i+j=r\\ k+\ell=r'}} x^{i+k}y^2\otimes x^\ell\otimes x^jy^2\\
&+\sum_{\substack{i+j=r\\ k+\ell=r'}} x^{i+k}y\otimes x^\ell y\otimes x^jy^2 + \sum_{\substack{i+j=r\\ k+\ell=r'}} x^{i+k}\otimes x^\ell y^2\otimes x^jy^2,
\end{aligned}$$

up to an overall constant and the appropriate power of ξ. Terms with $y^{k'}$, $k' \in \{0,1,2\}$, appearing in each position in the triple tensor product not found above either cancel in the sum or are not formed at all e.g. terms such as $x^i y^2 \otimes x^j y^2 \otimes x^k y^2$ cannot be formed.

By exchanging $k \leftrightarrow \ell$ and forming $i' = i + \ell$ and $j' = j - \ell$, and therefore $i' + j' = i + j$, we see that the above terms also feature in the analogous grouping in $(\mathrm{id} \otimes \Delta^S_{n,\xi}) \circ \Delta^S_{n,\xi}(1)$; all the other terms in the same grouping of the latter expression again either cancel or are not formed. Since all possible groupings are covered, the result is proven. □

Lemma. *(See [CT26a, Lemma 6A.4]). Given the color basis, we have the following:*

(i) $p_k \cdot p_m = \delta_{km} p_m$, *and hence* $E_{k,0} \cdot E_{m,0} = \delta_{km} E_{m,0}$ *and* $E_{k,\pm} \cdot E_{m,\pm} = \delta_{km} E_{m,\pm}$, *while all other products between color basis elements are zero;*

(ii) $\Delta_{n,1}(E_{k,0}) = -\frac{1}{3} f_{n,k} E_{k,0} \otimes E_{k,0}$ *and* $\Delta_{n,1}(E_{k,\pm}) = \frac{2}{3} f_{n,k} E_{k,\pm} \otimes E_{k,\pm}$;

(iii) $m_{n,1}(E_{k,0}) = u g_{n,k} E_{k,0}$ *and* $m_{n,1}(E_{k,\pm}) = (u+w) g_{n,k} E_{k,\pm}$,

where

$$f_{n,k} = \begin{cases} 1 & n \text{ even} \\ \omega^k & n \text{ odd} \end{cases}, \qquad g_{n,k} = \begin{cases} \omega^{kn/2} & n \text{ even} \\ \omega^{k(n+1)/2} & n \text{ odd} \end{cases}.$$

In other words, $\partial_{n,1}$ *is diagonal in the color basis.*

Proof. (i) The calcuation is straightfoward using the following result:

$$p_k \, x^{n-\ell} = \omega^{-k\ell} p_k$$

for all $\ell = 1, \dots, n$.

For (ii) and (iii), let us use the fact that $\xi = 1$ to simplify the expression for $\Delta_{n,1}$:

$$\begin{aligned} \Delta_{n,1}(v) = &- \frac{1}{3n} \sum_{i+j \equiv c_n \bmod n} (v \cdot x^i) \otimes x^j \\ &+ \frac{1}{3n} \sum_{i+j \equiv (c_n - 1) \bmod n} (v \cdot x^i y^2) \otimes x^j + (v \cdot x^i y) \otimes x^j y + (v \cdot x^i) \otimes x^j y^2. \end{aligned}$$

for $c_n = \begin{cases} 0 & n \text{ even} \\ 1 & n \text{ odd} \end{cases}$.

We now show that $\Delta_{n,1}(E_{k,0}) = -\frac{1}{3} f_{n,k} E_{k,0} \otimes E_{k,0}$. The other cases involving $E_{k,\pm}$ proceed similarly.

Firstly,

$$\begin{aligned} E_{k,0} \otimes E_{k,0} &= p_k(1 - x^{n-1}y^2) \otimes p_k(1 - x^{n-1}y^2) \\ &= p_k \otimes p_k - \omega^{-k} p_k \otimes p_k y^2 - \omega^{-k} p_k y^2 \otimes p_k + \omega^{-2k} p_k y^2 \otimes p_k y^2 \\ &= \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s)} x^r \otimes x^s - \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s+1)} x^r \otimes x^s y^2 \\ &- \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s+1)} x^r y^2 \otimes x^s + \frac{1}{n^2} \sum_{r,s=0}^{n-1} \omega^{-k(r+s+2)} x^r y^2 \otimes x^s y^2 \\ &= \frac{1}{n^2} \sum_{\substack{\ell,r,s=0 \\ r+s \equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r \otimes x^s - \frac{1}{n^2} \sum_{\substack{\ell,r,s=0 \\ r+s+1 \equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} (x^r \otimes x^s y^2 + x^r y^2 \otimes x^s) \\ &+ \frac{1}{n^2} \sum_{\substack{\ell,r,s=0 \\ r+s+2 \equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r y^2 \otimes x^s y^2. \end{aligned}$$

On the other hand,

$$\Delta_{n,1}(E_{k,0}) = \Delta_{n,1}(p_k) - \omega^{-k} \Delta_{n,1}(p_k y^2)$$

$$= -\frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv c_n \bmod n}}^{n-1} \omega^{-k\ell} x^{\ell+i} \otimes x^j + \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k\ell} x^{\ell+i} y^2 \otimes x^j$$

$$+ \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k\ell} (x^{\ell+i} y \otimes x^j y + x^{\ell+i} \otimes x^j y^2)$$

$$+ \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv c_n \bmod n}}^{n-1} \omega^{-k(\ell+1)} x^{\ell+i} y^2 \otimes x^j - \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k(\ell+1)} x^{\ell+i+1} y^2 \otimes x^j$$

$$- \frac{1}{3n^2} \sum_{\substack{\ell,i,j=0 \\ i+j\equiv (c_n-1) \bmod n}}^{n-1} \omega^{-k(\ell+1)} (x^{\ell+i+1} y \otimes x^i y + x^{\ell+i} y^2 \otimes x^j y^2)$$

$$= -\frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r \otimes x^s + \frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s+1\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r y^2 \otimes x^s$$

$$+ \frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s+1\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r \otimes x^s y^2 - \frac{1}{3n^2} f_{n,k} \sum_{\substack{\ell,r,s=0 \\ r+s+2\equiv \ell \bmod n}}^{n-1} \omega^{-k\ell} x^r y^2 \otimes x^s y^2,$$

and so the result follows.

Finally, (iv) and (v) follow straightforwardly. For example,

$$\begin{aligned} m_{n,1}(E_{k,0}) &= m_{n,1}(p_k) - \omega^{-k} m_{n,1}(p_k y^2) \\ &= u p_k x^{n/2} + w p_k x^{(n-2)/2} y^2 - \omega^{-k} u p_k x^{n/2} y^2 - \omega^{-k} w p_k x^{(n-2)/2+1} y^2 \\ &= u p_k x^{n/2} - \omega^{-k} u p_k x^{n/2} y^2 \\ &= u\omega^{kn/2} (p_k - \omega^{-k} p_k y^2) = u\omega^{kn/2} E_{k,0}, \end{aligned}$$

for n even, and the rest are similar. □

Theorem. *(See [CT26a, Theorem 7B.9]). For the complete graph K_4,*

$$\begin{aligned} Mh_{n,1}(K_4) = n\big(n^3(40b+41) - 6n^2(13b+14) + 11n(4b+5) - 6(b+2)\big)t^4 \\ + n\big(n^2(13b+14) - 3n(4b+5) + 2(b+2)\big)t^2, \end{aligned}$$

and for the complete bipartite graph $K_{3,2}$,

$$Mh_{n,1}(K_{3,2}) = n\big(n^2(13b+14) - 3n(4b+5) + 2(b+2)\big)(t^6 + 3t^4 + 3t^2 + 1).$$

Proof. We begin by inspecting the circle counts associated with the hypercube of states of the blowup of K_4, obtained using the code described in [CT26a, CT26b]:

```
circle counts by homological degree: [[2], [1, 2, 1, 1, 2, 1], [1, 2, 2, 1, 2, 1, 1,
    3, 1, 2, 1, 2, 1, 2, 1], [2, 1, 2, 1, 3, 1, 3, 2, 3, 1, 1, 2, 1, 2, 2, 2, 1, 3,
    2, 1], [2, 1, 2, 1, 2, 2, 2, 4, 2, 2, 2, 2, 1, 1, 2], [1, 3, 1, 1, 3, 1], [2]]
```

The maximum number of circles of any state is 4, and so by using [CT26a, Remark 7B.2], we only need to solve for the coefficients of a degree four polynomial in n without a constant term,

$$c_1 n^4 + c_2 n^3 + c_3 n^2 + c_4 n,$$

for each degree of t up to six and b up to one, using the data from Table 1 for K_4 up to $n = 4$. The only nontrivial systems of equations that require solving are

$$\begin{aligned} \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 &= 0 \\ 16\alpha_1 + 8\alpha_2 + 4\alpha_3 + 2\alpha_4 &= 180 \\ 81\alpha_1 + 27\alpha_2 + 9\alpha_3 + 3\alpha_4 &= 1512 \\ 256\alpha_1 + 64\alpha_2 + 16\alpha_3 + 4\alpha_4 &= 5928 \end{aligned} \qquad \begin{aligned} \beta_1 + \beta_2 + \beta_3 + \beta_4 &= 0 \\ 16\beta_1 + 8\beta_2 + 4\beta_3 + 2\beta_4 &= 180 \\ 81\beta_1 + 27\beta_2 + 9\beta_3 + 3\beta_4 &= 1512 \\ 256\beta_1 + 64\beta_2 + 16\beta_3 + 4\beta_4 &= 5952 \end{aligned}$$

4	0	$4(15b+18)(3t^2+1)$	$24(247b+248)t^4+24(27b+28)t^2$
3	0	$3(11b+13)(3t^2+1)$	$504(3b+3)t^4+3(83b+85)t^2$
2	0	$2(7b+8)(3t^2+1)$	$20(3b+3)(3t^4+t^2)$
1	0	$(3b+3)(3t^2+1)$	$(3b+3)t^2$
n / m	2	3	4

TABLE 1. $Mh_{n,1}(K_m)$ for a range of m, n.

4	0	0	$4(15b+18)(t^4+6t^2+1)$	$24(27b+28)(t^6+3t^4+3t^2+1)$
3	0	0	$3(11b+13)(t^4+6t^2+1)$	$3(83b+85)(t^6+3t^4+3t^2+1)$
2	0	0	$2(7b+8)(t^4+6t^2+1)$	$20(3b+3)(t^6+3t^4+3t^2+1)$
1	0	0	$(3b+3)(t^4+6t^2+1)$	$(3b+3)(t^6+3t^4+3t^2+1)$
n / (m,p)	$(1,1)$	$(2,1)$	$(2,2)$	$(3,2)$

TABLE 2. $Mh_{n,1}(K_{m,p})$ for a range of m, p, n.

associated with degree one and zero in b respectively for degree four in t, and similarly for degree two in t:

$$\begin{aligned} \delta_1+\delta_2+\delta_3+\delta_4 &= 3 & \gamma_1+\gamma_2+\gamma_3+\gamma_4 &= 3 \\ 16\delta_1+8\delta_2+4\delta_3+2\delta_4 &= 60 & 16\gamma_1+8\gamma_2+4\gamma_3+2\gamma_4 &= 60 \\ 81\delta_1+27\delta_2+9\delta_3+3\delta_4 &= 249 & 81\gamma_1+27\gamma_2+9\gamma_3+3\gamma_4 &= 255 \\ 256\delta_1+64\delta_2+16\delta_3+4\delta_4 &= 648 & 256\gamma_1+64\gamma_2+16\gamma_3+4\gamma_4 &= 672 \end{aligned}$$

The reader is encouraged to check that the solutions are given by

$$(1) \qquad \begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\\ \alpha_4\end{pmatrix} = \begin{pmatrix}40\\ -78\\ 44\\ -6\end{pmatrix}, \quad \begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\\ \beta_4\end{pmatrix} = \begin{pmatrix}41\\ -84\\ 55\\ -12\end{pmatrix}, \quad \begin{pmatrix}\delta_1\\ \delta_2\\ \delta_3\\ \delta_4\end{pmatrix} = \begin{pmatrix}0\\ 13\\ -12\\ 2\end{pmatrix}, \quad \begin{pmatrix}\gamma_1\\ \gamma_2\\ \gamma_3\\ \gamma_4\end{pmatrix} = \begin{pmatrix}0\\ 14\\ -15\\ 4\end{pmatrix}.$$

For $K_{3,2}$, the circle counts associated with the hypercube of states of the blowup are given by:

```
circle counts by homological degree: [[3], [2, 2, 2, 2, 2, 2], [3, 1, 1, 1, 1, 1, 1,
    1, 1, 3, 1, 1, 1, 1, 3], [2, 2, 2, 2, 2, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 2, 2, 2,
    2, 2], [3, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, 1, 1, 1, 3], [2, 2, 2, 2, 2, 2], [3]]
```

As above, we only need to solve for the coefficients of a degree three polynomial in n without a constant term, i.e. of the form $c_1n^3+c_2n^2+c_3n$, for each degree of t up to six and b up to one, using the data from Table 2 for $K_{3,2}$ up to $n=3$. Comparing with the data for K_4, it is straightforward to see that in fact the solutions for the coefficients δ_i, γ_i, $i=1,\dots,4$ from Equation 1 can be reused to solve the nontrivial systems of equations for $K_{3,2}$, up to overall constant factors. □

REFERENCES

[CT26a] D. W. Collison and D. Tubbenhauer. Categorifying the Penrose polynomial using nonorientable surfaces. 2026. https://arxiv.org/abs/2607.01632.

[CT26b] D. W. Collison and D. Tubbenhauer. Code, data and more for "Categorifying the Penrose polynomial using nonorientable surfaces". 2026. https://github.com/dtubbenhauer/graphhomology.

D.W.C.: THE UNIVERSITY OF SYDNEY, SCHOOL OF MATHEMATICS AND STATISTICS F07, NSW 2006, AUSTRALIA
Email address: daniel.collison@sydney.edu.au

D.T.: THE UNIVERSITY OF SYDNEY, SCHOOL OF MATHEMATICS AND STATISTICS F07, OFFICE CARSLAW 827, NSW 2006, AUSTRALIA, WWW.DTUBBENHAUER.COM, HTTPS://ORCID.ORG/0000-0001-7265-5047
Email address: daniel.tubbenhauer@sydney.edu.au